Smart Energy Management and Load Optimization for Commercial Microgrid Systems

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***Abstract*—The increasing penetration of renewable energy sources in commercial buildings has amplified the need for intelligent energy management systems capable of coordinating distributed generation, storage, and heterogeneous loads. Commercial microgrids integrating solar photovoltaic systems, battery storage, and grid supply often suffer from inefficient energy utilization due to fragmented monitoring platforms, limited automation, and lack of real-time decision-making. High-consumption loads such as Heating, Ventilation, and Air Conditioning (HVAC) systems and split air conditioners dominate energy usage in such environments, making their coordinated control critical.**

This paper presents a comprehensive smart energy management and load optimization framework for commercial hybrid microgrids. The proposed system enables real-time data acquisition from legacy, non-smart solar inverters using industry-standard communication protocols and retrofits conventional HVAC and split AC systems for centralized monitoring and control. A custom integration built on the Home Assistant platform provides a unified interface for visualization, automation, and user interaction. Furthermore, an AI-assisted decision-making framework based on Retrieval-Augmented Generation (RAG) is implemented locally to analyze real-time energy data and generate actionable load control recommendations.

The system is validated through hardware deployment in a real commercial environment. Experimental results demonstrate improved renewable energy utilization, reduced grid dependency, and effective prioritization of high-energy loads, while maintaining low computational complexity and practical deployability. The proposed framework offers a scalable and cost-effective solution for small-to-medium commercial microgrids.

***Index Terms*—Commercial microgrids, energy management systems, load optimization, HVAC control, Home Assistant, Modbus RTU, retrieval-augmented generation**

I. Introduction

The global energy sector is undergoing a fundamental transition driven by increasing energy demand, climate concerns, and the rapid adoption of renewable energy technologies. Solar photovoltaic (PV) systems have become a cornerstone of distributed energy generation due to their modularity, declining costs, and widespread availability. In commercial buildings, PV systems are increasingly deployed alongside battery storage and grid interconnection, forming hybrid microgrids capable of improving energy resilience and reducing operational costs.

Despite these advancements, effective energy utilization in commercial microgrids remains a significant challenge. Commercial buildings exhibit highly variable and demand-intensive load profiles dominated by HVAC systems, split air conditioners, lighting, and other critical infrastructure. These loads are often operated independently of real-time energy availability, resulting in inefficient grid usage, underutilization of renewable generation, and increased electricity expenses.

Most commercial installations rely on fragmented energy monitoring and control solutions. Solar inverters, HVAC systems, and smart switches are typically managed through vendor-specific applications, preventing holistic optimization. Advanced energy management solutions proposed in literature often assume IoT-native infrastructure and standardized interfaces, which are rarely available in legacy commercial environments.

This work addresses these challenges by presenting a practical, hardware-validated smart energy management and load optimization system tailored for commercial microgrids. The proposed framework emphasizes retrofitting existing infrastructure, real-time monitoring, intelligent load prioritization, and user-centric control through an open-source platform.

II. Background and Related Work

A. Commercial Microgrid Architectures

Microgrids are localized energy systems that integrate distributed energy resources, storage, and controllable loads within defined boundaries. Solar-based microgrids are commonly categorized into off-grid, on-grid, and hybrid configurations. Hybrid microgrids, which combine PV generation,

battery storage, and grid connectivity, offer superior reliability and flexibility for commercial applications but require intelligent coordination to operate efficiently.

B. Energy Management and Load Optimization Approaches

Energy management in microgrids has been extensively studied using optimization-based and AI-driven techniques. Multi-objective optimization methods such as adaptive genetic algorithms have been applied to minimize cost, emissions, and power losses while handling discrete variables [1]. Commercial simulators provide user-friendly platforms for microgrid planning and ROI estimation [2]. Recent AI-based frameworks using Greylag Goose Optimization combined with graph convolutional networks achieve high load forecasting accuracy in smart building microgrids [3]. Bio-inspired algorithms for demand-side management in residential smart grids schedule shiftable loads to reduce peaks and costs [4].

C. Limitations of Existing Solutions

Most existing solutions are simulation-centric and lack real-world hardware validation. They often assume IoT-ready devices, standardized APIs, and high-resolution data streams, which are uncommon in legacy commercial setups. Furthermore, many approaches treat loads in aggregate, ignoring direct control of dominant energy consumers such as HVAC and split AC systems. User interaction and manual override mechanisms are frequently overlooked, reducing real-world usability.

D. Contribution of This Work

The proposed system distinguishes itself by focusing on deployability, retrofitting, and practical control. It integrates non-smart inverters via Modbus RTU, enables direct control of HVAC and split AC systems, and provides a unified interface using Home Assistant. The AI-assisted decision-making framework prioritizes actionable recommendations with low computational overhead.

III. System Overview

The proposed system is organized into three functional layers: data acquisition, decision-making, and load control. This layered architecture enables modular development, scalability, and ease of integration.

A. Data Acquisition Layer

This layer collects real-time energy data from the solar microgrid. Legacy solar inverters are interfaced using RS232/RS485 with Modbus RTU. A custom ESP32-based hardware node acts as a data logger, periodically polling inverter registers.

B. Decision-Making Layer

The decision-making layer processes real-time data using rule-based automation in Home Assistant and an AI-assisted Retrieval-Augmented Generation module to maximize renewable utilization while preserving comfort.

C. Load Control Layer

The load control layer interfaces with HVAC systems via custom wired controllers and split ACs via infrared blasters. Smart switches with native Home Assistant support are also incorporated.

IV. Hardware Implementation

A. Inverter Data Acquisition Hardware

A custom ESP32 module with RS232/RS485-to-UART converters connects directly to the inverter and transmits data wirelessly to Home Assistant.

B. HVAC Control Hardware

Commercial HVAC controllers were reverse-engineered. Custom ESP32-based units with touch interfaces enable both manual and remote control without replacing existing equipment.

Fig. 1. Custom ESP32-based HVAC controller hardware with touch interface for manual operation and wireless Home Assistant integration.

C. Split AC Control Hardware

Split AC units are controlled using compact infrared blaster modules that transmit manufacturer-specific IR codes, converting legacy units into remotely controllable loads.

Fig. 2. Infrared blaster module for split AC control. Multiple transmitters allow simultaneous management of several units in a single room.

V. Software Architecture

A. Firmware for Smart Controllers

All custom devices run firmware that exposes them as discoverable entities over Wi-Fi, enabling seamless monitoring and control through Home Assistant.

B. Home Assistant Integration

A custom integration unifies inverter parameters, load states, and control commands into a single dashboard for visualization, automation, and historical logging.

VI. AI-Assisted Decision Making

A. Retrieval-Augmented Generation Framework

A proprietary rule database is embedded into a vector store. Real-time inverter data serves as a query for semantic retrieval of relevant decision contexts.

B. Local Large Language Model

A locally hosted LLM generates a concise system summary and recommended actions from the retrieved context and live data. Actions are presented in Home Assistant for one-click execution.

VII. Experimental Evaluation

The system was deployed in a commercial building equipped with a hybrid solar microgrid, HVAC units, and split ACs. Evaluation focused on communication reliability, system responsiveness, and energy utilization. The inverter data acquisition system demonstrated stable communication over extended operation. Automated load prioritization reduced grid import during peak solar generation periods and improved battery utilization. The system operated reliably under real-world conditions with minimal latency.

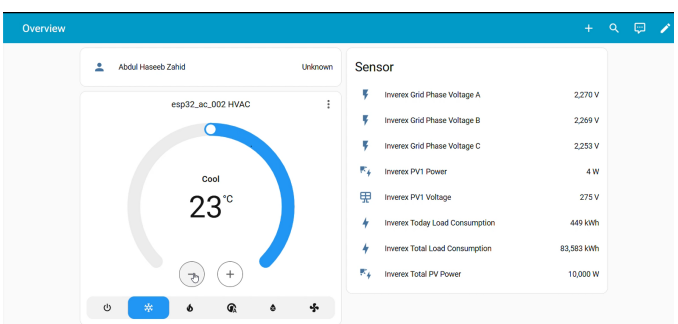

Fig. 3. The Home Assistant interface displaying real-time data from the inverter, along with HVAC and AC control interfaces, demonstrating the unified monitoring and control in the implemented system.

VIII. Discussion

The framework strikes a practical balance between intelligence and deployability. By focusing on direct load control rather than heavy forecasting, it achieves effective optimization with minimal computational resources. The current scope covers HVAC, split ACs, and smart switches; the modular architecture readily supports future addition of occupancy sensors and smart plugs.

IX. Conclusion and Future Work

This paper presented a deployable smart energy management system for commercial microgrids that retrofits legacy equipment and integrates AI-assisted decision making via Home Assistant. The hardware-validated solution improves renewable utilization and reduces operational costs while preserving user comfort. Future work will incorporate occupancy-aware scheduling, expand controllable loads, and conduct long-term field studies across multiple sites.

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